

Code-Layout, Readability and Reusability

Date	11 July 2026
Team ID	
Project Name	Rising waters - Flood Prediction using Machine Learning
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and tabs are used throughout the code to maintain consistency.	Yes	PEP-8 standards followed for Python code.
2	Proper File Structure	Files and folders are logically organized based on functionality.	Yes	Separate folders for templates, static, model, dataset and app.
3	Meaningful Variable Names	Variables reflect their purpose clearly and are easy to understand.	Yes	Descriptive variable names used across the project.
4	Function / Method Names	Functions and methods are descriptively named based on their purpose.	Yes	Clear and meaningful function names used.
5	Code Comments	Inline and block comments explain important logic and complex steps.	Partial	Comments added for major modules and important logic.
6	Modular Design	Code is split into reusable functions and modules for better maintainability.	Yes	Modules for preprocessing, model training, prediction and routes.
7	No Redundant Code	Duplicate or unused code is removed to keep the code clean.	Yes	Unused code removed and refactored.
8	Error Handling	Exceptions and errors are handled gracefully to prevent crashes.	Yes	Try-except blocks used for user input and prediction errors.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python (Pandas)	Training & Prediction	High

2	Machine Learning Model	Scikit-Learn	Prediction Engine	High
3	Prediction API	Flask	Web Application	High
4	Frontend UI	HTML,CSS,JavaScript	User Dashboard	Medium
5	Model Loader	Pickle(Python)	Prediction Module	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-structured project with modular folders.
Readability	5	Meaningful naming conventions and clean formatting.
Reusability	5	Modules reused across training and prediction workflows.
Documentation / Comments	4	Important sections documented; additional comments can improve maintainability
Overall Score	4.8/5	Clean, modular, readable, and reusable codebase suitable for future enhancements.